

Make a Climate Light

A tiny sensor turns raw timing into an unmistakable green success

Build one working layer at a time. Each new light unlocks the next.

Lesson	013 — component	Time	about 45 minutes
You need	Lessons 001–012	Board	Arduino Mega 2560
Main signal	D22	Visible result	RGB LED on D5–D7
Power	USB only	Stop	unplug USB

The one specimen decision

The normal kit path is an **identified, labeled DHT11 module**. The Elegoo listing authorizes that family; it does not identify the board revision, pin order, supply limit, current, or pull-up. Before power, read the markings on the module in your hand and its matching documentation. Continue only when they establish 5 V operation, +, DATA (often S), –, peak current, and whether a DATA pull-up is already fitted. The drawing uses those *names*, not an asserted left-to-right module order.

Before power: With USB unplugged, map the module’s printed +, S/DATA, and – labels to breadboard columns 8, 9, and 10; rearrange the three module leads if its physical order differs. Use one 330 Ω resistor for each LED die. If identity, supply rating, current, LED common cathode, or pull-up is uncertain, set that part aside. Stop for heat, odor, resets, or a warm jumper.

If the identified module has pull-up R_P , its worst DATA-low current is $5/R_P$; if it has none, fit the documented value from DATA to 5 V. The 330 Ω LED resistors keep ordinary documented indicator dies below 10 mA each when their minimum forward voltage is at least 1.8 V. Confirm the module’s peak current and keep module plus three LED channels below 35 mA.

1 Plate 1 — Find the real pins

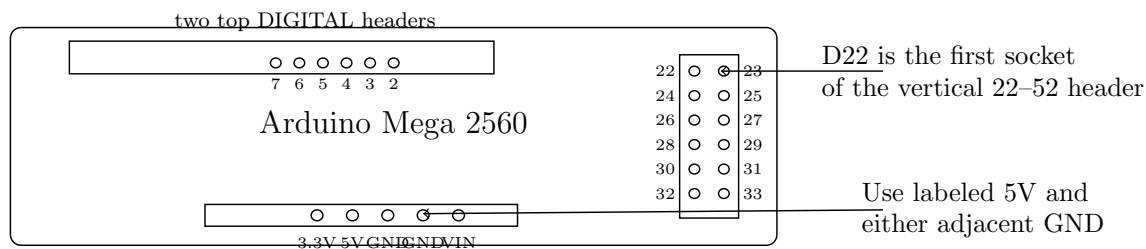


Figure 1: Top-view locator plate. The Mega’s D22 is the first socket of the paired 22–53 digital header, not socket 2 on the top header. The power sockets are the labeled 5V and GND positions.

Put the Mega beside the unpowered breadboard exactly as drawn. Find D22, D5, D6, D7, 5V, and GND with a fingertip before placing any wire.

You win when: you can point to all six Mega sockets without following a jumper.

2 Plate 2 — Build the whole climate light

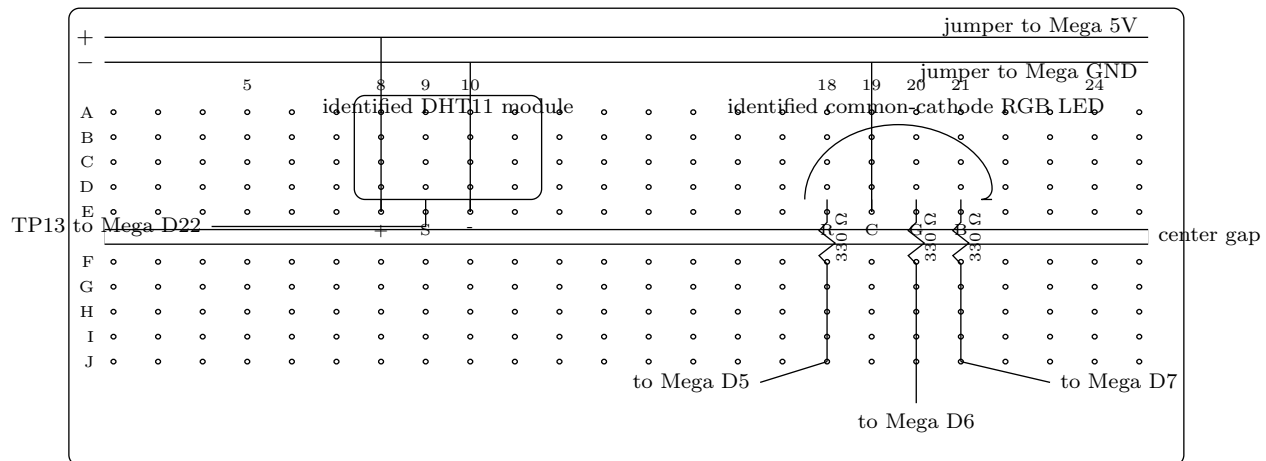


Figure 2: Literal breadboard plate. Numbered columns run left to right; A–E and F–J are separate five-hole groups divided by the center gap. Module names land at E8–E10. The RGB legs land at E18–E21; each anode crosses the gap through its own resistor, while common cathode E19 reaches the minus rail. Jumper endpoints are named on the plate.

From	To	What this means
Mega 5V	breadboard + rail	one 5 V supply rail
Mega GND	breadboard – rail	one common return rail
module + at E8	+ rail	VCC; only if documented for 5 V
module S/DATA at E9	Mega D22	DATA; this same group is TP13
module – at E10	– rail	GND
LED red E18	resistor E18–F18, then J18 to D5	red channel
LED common cathode E19	– rail	shared return
LED green E20	resistor E20–F20, then J20 to D6	green channel
LED blue E21	resistor E21–F21, then J21 to D7	blue channel

The pictured module body is deliberately generic. Rotate or jumper the identified module so its *named* leads reach E8 +, E9 DATA, and E10 –; do not copy an unverified physical order. Likewise, preserve the identified LED’s electrical lead names if its leg order is not R–C–G–B.

Inspect every endpoint, plug in USB, and upload. The complete first powered stage already has its payoff: blue pulses during the short start, then a steady valid color appears. There is no Serial ritual or worksheet.

3 Plate 3 — Watch the unlocks

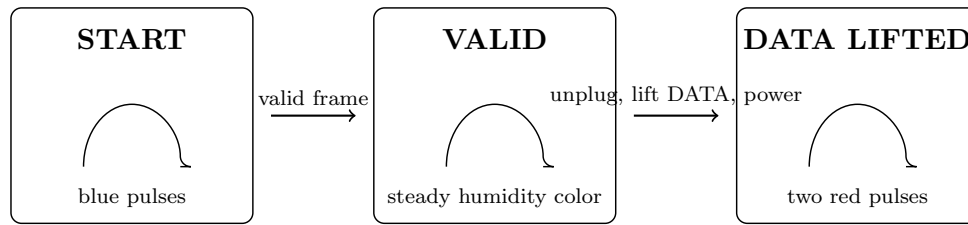


Figure 3: Visible progression. A valid frame automatically advances from the start pulse to a steady humidity color. The reversible missing-DATA surprise cannot reach a valid color; restoring the jumper restarts the progression.

You win when: blue pulses while the sensor starts, then the lamp settles on a steady green, cyan, or amber. That transition can happen only after a complete 40-bit frame passes timing, checksum, range, and freshness rules.

4 Play with the air

Cup your hands loosely around the sensor without touching it, then breathe gently across the vents from several inches away. The steady valid color moves from green (below 55% RH), through cyan (55–69.9% RH), to amber ($\geq 70\%$ RH). Move it back into room air and watch it return. Lesson 014 will reveal the numbers.

Now unplug USB, lift only the D22-to-DATA jumper, and restore USB. The sketch cannot fake success: two red pulses repeat because no bounded sensor reply arrives. Unplug USB, restore that one jumper, and power again. Blue then green is the recovery animation.

What the lamp does	What to fix next
No light	unplug USB; check LED type, common cathode, D5–D7, and all three resistors
Blue forever	unplug USB; recheck D22, module labels, 5V, GND, and the documented pull-up decision
Two red pulses	the sensor transaction timed out; check the DATA path
Three red pulses	a frame arrived but failed checksum or range validation; shorten jumpers and inspect supply and specimen identity
Amber	the last good observation became old; reset and watch for a new frame
Steady green/cyan/amber	a current frame passed every software rule; color shows its humidity band, not calibrated accuracy

5 What the code is doing

The sketch reads in the same order as the build:

```

const adk::Status      status      = observeClimate    (now);
const adk::ClimateSample observation = decideClimateState (now);

if ((!status.ok () &&
    observation.state == adk::ClimateSampleState::Unavailable) ||
    !showClimateState (observation, now))
{
    haltClimateCircuit ();
}
  
```

`Dht11Sensor` owns D22. It drives DATA low only for the request and then releases it to input; every edge wait has a deadline. It accepts a frame only after all five bytes arrive, their checksum matches, and the decoded values are in range. `sample(now, freshFor)` can later mark an old good frame stale without touching the wire. Shutdown releases D22 and turns the RGB LED off.

The repository quietly replays good frames, timeouts, malformed pulses, checksum faults, timing rollover, initialization failures, restart, and shutdown. Those automated checks protect the lesson; they are not chores for the learner.

6 Next unlock

You now have a circuit that distinguishes “a sensor said something” from “a complete, current frame passed every rule.” Keep this exact build intact. Lesson 014 gives the observations a character display, and lesson 015 turns both components into a small environmental station.